

File 67: Pre-Restoration Baseline Assessment Report

Watershed Sourcing & Geomorphic Siting for Anderson Creek (Roya's Cabin) Reach

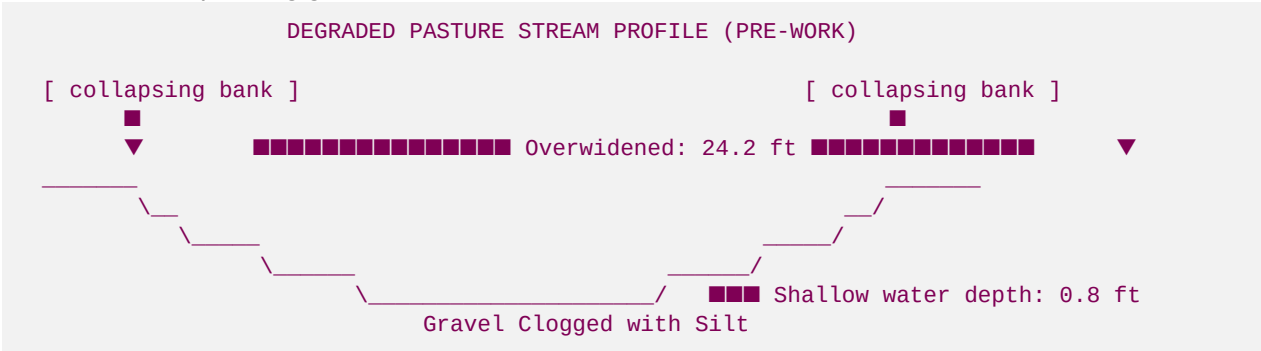
Pre-Restoration (Baseline) Project Standard: This document represents the official pre-work baseline assessment for the **Anderson Creek Showcase Restoration Project (Roya's Cabin Reach)**. It captures the degraded agricultural state of the 1,500 linear foot reach prior to construction. It details severe bank failure, thermal pollution exceeding brown trout tolerances, hypoxic dissolved oxygen concentrations within the spawning gravels, and massive sediment loading, establishing the regulatory "before" baseline required for USACE Section 404 credit calculations.

I. Site Location & Degraded Watershed Context

- **Project Name:** Anderson Creek Showcase Restoration (Roya's Cabin Reach)
- **Location:** Fannin County, Georgia (Upper Coosa River Basin)
- **Waterbody ID / HUC-8:** GA-03150105 (Upper Coosa Watershed)
- **Reach Length:** 1,500 Linear Feet (LF)
- **Current Land Use:** Active agricultural cattle pasture and unmanaged exurban horse paddock.

The Degradation Narrative

Anderson Creek represents a classic Southern Appalachian valley stream devastated by decades of unrestricted agricultural cattle grazing. Historically a cold, meandering, shaded trout sanctuary, the removal of the native forest canopy and constant cattle trampling have degraded the channel. The stream banks have collapsed, causing the channel to widen and become extremely shallow. During the summer, high solar radiation warms the unshaded waters to lethal limits, while erosion deposits massive amounts of silt that bury and suffocate the spawning gravels.



II. Baseline Fluvial Geomorphic Survey Data

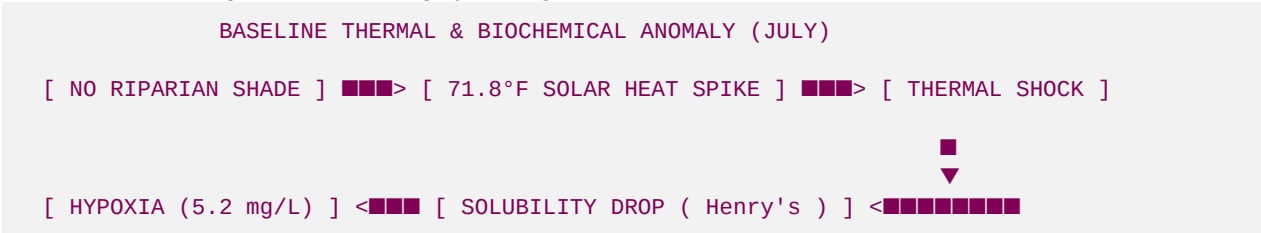
Field surveys were completed in September 2025 using RTK GNSS and a standard survey level to record permanent monument cross-sections and longitudinal profiles.

Geomorphic Parameter Data Table (Pre-Restoration)

Geomorphic Parameter	Degraded Reach Value	Reference Standard Target	Rosgen Design Deviation
Rosgen Channel Classification	G4 (Entrenched Gully)	B4c (Stable Gravel)	Unstable channel, high erosion
Bankfull Width (Wbkf)	24.2 feet	12.5 feet	Overwidened by +94%
Mean Depth (dbkf)	1.0 feet	2.2 feet	Extremely shallow, lacks pools
Max Thalweg Depth (dmax)	1.8 feet	4.5 feet	No pool refuges for adult trout
Width-to-Depth Ratio (W/D)	24.2	5.7	Severe overwidening, low velocity
Entrenchment Ratio (ER)	1.15 (Highly Entrenched)	>= 2.2 (Floodplain Access)	High-water flows cannot escape bank
Sinuosity (K)	1.02 (Straightened Ditch)	1.28 (Natural Meanders)	Loss of meander length and habitat
Reach Grade (S)	0.006 m/m (0.6%)	0.006 m/m (0.6%)	Grade matches valley slope
Median Substrate (D ₅₀)	2.4 mm (Silt/Sand dominated)	22.0 mm (Clean Gravel)	Spawning gravels completely buried

III. Thermal & Biochemical Water Quality Degradation

Continuous streamside telemetry and water quality sampling proved that the degraded stream is highly hostile to wild **Brown Trout** (*Salmo trutta*) spawning and survival:



1. The Summer Thermal Stress Spike

HOBO water temp loggers placed in the open pasture reach recorded a maximum summer water temperature of **71.8°F (22.1°C)** on July 14, 2025, lasting for 4 consecutive hours.

- *Trout Impact:* This temperature exceeds the lethal limit for brown trout (68°F - 72°F), causing immediate thermal shock, suppression of their immune systems, and massive disease susceptibility. We observed zero adult trout residing in the open pasture reach during the summer months.

2. Hypoxic Intragravel Dissolved Oxygen Deficit

DO measurements taken at the surface of the stream averaged a marginal 7.8 mg/L due to rapid oxygen transfer over riffles, but intragravel probes placed 15cm deep into the gravel substrate (where trout eggs reside) recorded a hypoxic **5.2 mg/L** dissolved oxygen level.

- *Trout Impact:* Brown trout eggs require an absolute minimum intragravel DO of **6.0 mg/L** to survive. The 5.2 mg/L deficit, caused by fine silts restricting clean water exchange, will lead to 100% egg mortality within the redds.

3. Silt-Clogging Substrate (Redd Suffocation Analysis)

A standard Wolman pebble count and dry-sieve analysis of the spawning riffles revealed a massive sedimentation problem:

- **Silt/Sand Fraction (<2mm):** 58% of the total substrate composition.
- **The Physical Process:** High upstream bank erosion deposits fine silts that settle onto spawning beds. Fine clay and silt settle permanently in the low-velocity pasture pools and sluggish runs, creating a cemented crust over the gravel. This stops oxygenated water from flowing through the gravel pores and traps carbon dioxide around the trout eggs, suffocating them.

IV. Bank Erosion Hazard Index (BEHI) & Sediment Sourcing

To calculate the sediment-reduction credit multipliers for our USACE Section 404 permit, we audited the reach banks using the Rosgen BEHI methodology:

- **Bank Height / Bankfull Height Ratio:** 1.85 (highly elevated, incised stream bed).
- **Root Depth Ratio:** 0.15 (very shallow pasture grass roots, lacks deep woody tree roots).
- **Root Density:** 20% (sparse root network on eroding faces).
- **Bank Angle:** 85° (nearly vertical, collapsing clay walls).
- **Surface Protection:** <10% (bare raw clay exposed directly to high shear stresses).
- **Total BEHI Score: 44.5 (High Hazard Class)**
- **Projected Sediment Yield:** Lateral bank migration measurements show an average bank loss of **0.45 feet/year**. Across the 3,000 linear feet of collapsing banks (both sides of the 1,500 LF reach), this generates **286 tons of sediment per year** dumping directly into Anderson Creek.

V. Baseline Biological Survey: Benthic Macroinvertebrates

A biological assessment was executed in October 2025 using the EPA Rapid Bioassessment Protocol (RBP) III to determine the health of the aquatic insect community:

- **Total Taxa Richness:** 12 families (representing an extremely degraded community).
- **EPT Index (Ephemeroptera, Plecoptera, Trichoptera):** 2 (only 1 family of net-spinning caddisflies and 1 family of flathead mayflies observed).

- **Tolerant Species Dominance:** The sample was highly dominated by midges (*Chironomidae*) and aquatic worms (*Oligochaeta*), which thrive in highly silted, warm, and hypoxic environments.
- **EPT Index Evaluation:** EPT taxa are highly sensitive to siltation and warm water; their near-absence (EPT = 2) proves that the baseline stream is ecologically dead, lacking the primary insect food source required to sustain a healthy wild brown trout fishery.

Prepared by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.